

SWAPCELL · DESIGN DECISION RECORD

SwapCell TRL 2 review decisions and interface v0.3

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SWC-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/swapcell		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 14, 16 and 18 decided by Amish on 2026-10-02
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and issue SwapCell interface v0.3

0001: TRL 2 review decisions and interface v0.3

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 13; items 15, 17, 19 and 20 accepted later on 2026-09-25 in SWC-DDR-002); items 14, 16 and 18 decided by Amish on 2026-10-02 (SWC-DEC-001)

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-24) listed eleven items as "Proposed, awaiting Amish", most with a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." He also approved three cross-cutting additions to the SwapCell interface, raised by the PowerBox, StepGen, SunSpoke, WaterWalker and CargoMule designs, and a portfolio pricing rule for shared packs.

This record lists what that instruction decides, and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (2026-09-24) and SWC-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1 lists the decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation" unless the row says otherwise.

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Cell configuration	Decided by Amish, 2026-09-25: go with recommendation. 13S2P of 5 Ah 21700 cells as the reference pack; 13S4P "double" pack only as a later variant in a longer envelope	SWC-PRC-001 v0.3, SWC-REQ-001 v0.3
2	Housing	Decided by Amish, 2026-09-25: go with recommendation. Folded aluminium tray with a printed flame-retardant lid and caps	SWC-PRC-001 v0.3, SWC-DWG-002
3	Interface v0.2 as written	Decided by Amish, 2026-09-25: go with recommendation. Envelope, 2 power plus 6 signal contacts with a last-mate interlock, CAN 2.0B at 250 kbit/s with 11-bit IDs. Superseded in the same session by v0.3 (items 11 to 13), which keeps every v0.2 feature	SWC-PRC-001 v0.3, interface section
4	Message set	Decided by Amish, 2026-09-25: go with recommendation. Open SwapCell profile now, with an EnergyBus (CiA 454) gateway study	SWC-PRC-001 v0.3. The gateway study is not complete; see item 18
5	Legacy mode for non-CAN vehicles	Decided by Amish, 2026-09-25: go with recommendation. Discharge only, fixed at 15 A, never charge	SWC-PRC-001 v0.3 behavior rules, SWC-REQ-001 R13
6	State-of-health log	Decided by Amish, 2026-09-25: go with recommendation. Log stored in the pack and read by any dock; no cloud dependency	SWC-REQ-001 R12
7	Dock charger	Decided by Amish, 2026-09-25: go with recommendation. Dock switches a certified 5 A charger; no custom mains electronics	SWC-PRC-001 v0.3
8	Fleet charge mode	Decided by Amish, 2026-09-25: go with recommendation. 4.1 V per cell (53.3 V) mode, selected by the dock	SWC-PRC-001 v0.3, SWC-REQ-001 R9
9	Interface governance	Decided by Amish, 2026-09-25: go with recommendation. A versioned interface section in this repo; changes approved by Amish; dependent designs build to it and flag conflicts to this repo	SWC-PRC-001 v0.3
10	Budget	Decided by Amish, 2026-09-25: go with recommendation. Keep <code>budget_usd: 700</code> for TRL 3; it covers one pack and one dock	<code>project.yaml</code> , SWC-REQ-001 R16
11	Interface addition W: wake for hosts without CAN or a wake supply	Decided by Amish, 2026-09-25 (cross-cutting approval): add a wake method. Implemented in v0.3 as wake on a coded INTERLOCK loop, with a manual wake button; the WAKE pin stays optional	SWC-PRC-001 v0.3, SWC-REQ-001 R13
12	Interface addition C: charge-while-discharging mode	Decided by Amish, 2026-09-25 (cross-cutting approval): add the mode. Implemented in v0.3 as requested mode 4 (charge-discharge) for dock, station and vehicle hosts	SWC-PRC-001 v0.3, SWC-REQ-001 R14
13	Interface addition V: latch vibration rating for vehicles	Decided by Amish, 2026-09-25 (cross-cutting approval): add a rating. Implemented in v0.3 as latch class V1 for vehicle receivers and class D for gravity docks	SWC-PRC-001 v0.3, SWC-REQ-001 R15

Two further approvals from the same instruction are recorded here:

- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting): a SwapCell pack is priced in this repo only and excluded from each dependent kit budget. Recorded in SWC-REQ-001 R16 and `bom/bom-notes.md`.
- **Solar DC dock variant.** Decided by Amish, 2026-09-25: go with recommendation (SWC-PRC-001 v0.2 open questions). A second dock variant with a 48 V DC solar input, via PowerBox, follows the mains dock. It is not designed at TRL 3.

Items that remain open

These had no recommendation, or the recommendation was to decide later, so they stayed **Proposed, awaiting Amish** in v0.1. On 2026-09-25 Amish accepted all remaining recommendations (SWC-DDR-002); the status column below is updated. Items with no recommendation stay open.

Table 2. Open items.

#	Item	Status
14	First co-design partner (repair workshop, delivery fleet or rural e-bike cooperative)	Decided by Amish, 2026-10-02: kept open under the portfolio rule, but an urban cargo-bike or e-bike delivery fleet is approached first, with a repair workshop second
15	Build budget for a second pack before any build (about \$900 at TRL 2; about \$973 for two packs and one dock with the TRL 3 prices)	Decided by Amish, 2026-09-25: go with recommendation. Keep \$700 for TRL 3 and set the build budget before any build. On hold: TRL 4 is on hold (SWC-DDR-002)
16	LFP variant (16S) sharing the envelope and message set	Decided by Amish, 2026-10-02: not added now; a chemistry code and a coding key are reserved so an LFP pack can never take an NMC charge or the reverse
17	Connector family	Decided by Amish, 2026-09-25: go with recommendation. Custom keyed shroud with commercial high-current socket contacts and potted signal contacts (SWC-DDR-002)
18	EnergyBus gateway detail	Decided by Amish, 2026-10-02: deferred; the native SwapCell profile stays the reference and CiA 454 is read only when a partner needs EnergyBus. Previously: The message-by-message mapping needs the CiA 454 specification, which is distributed through a membership organization and was not read in this session
19	Values chosen inside the approved additions W, C and V (10 k Ω coded interlock, 100 μ A sleep limit, 25 g shock level, 330 N receiver preload, charge-FET fallback rule)	Decided by Amish, 2026-09-25: go with recommendation. Values confirmed as proposed (SWC-DDR-002)
20	R3 thermal rating: keep 20 A continuous with temperature derating, or derate the label to 15 A	Decided by Amish, 2026-09-25: go with recommendation. Keep 20 A at 25 °C with the PACK_LIMITS derating; R3 restated (SWC-DDR-002)

■ Consequences

- SwapCell interface **v0.3** is issued in SWC-PRC-001 v0.3. It keeps every v0.2 signal, message and rule, so a v0.2 host that drives WAKE and sends a heartbeat needs only the change in the next point; v0.3 hosts can also wake a pack with no supply of their own.
- Every v0.3 receiver must fit a 10 kΩ coding resistor in its INTERLOCK loop (a direct link to SGND is no longer valid). This is the one change a v0.2 receiver design must make.
- Dependent designs (PowerBox, SunSpoke, StepGen, WaterWalker, CargoMule and others) cite these as "SwapCell interface v0.3" items W, C and V.
- SWC-PRB-001, SWC-PRC-001 and SWC-REQ-001 move to v0.3. TRL is set to 3. No TRL 4 work is started.